

What Is Hallucination in LLM?

An AI hallucination refers to an output generated by an AI model that deviates from reality or lacks a factual basis.

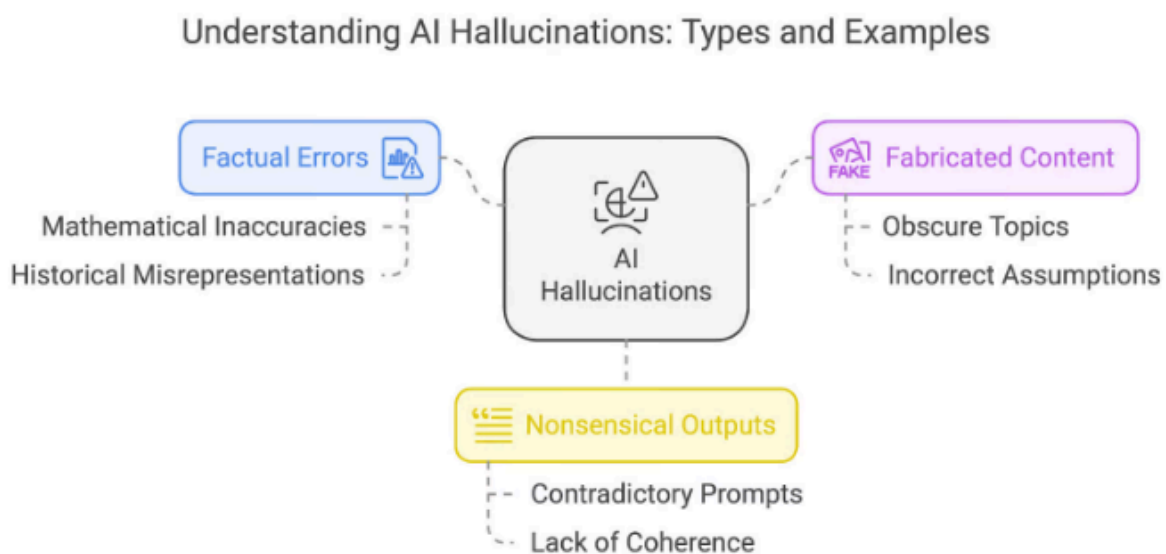
In simpler terms, this is an instance where the model provides an incorrect answer, fabricates a story, or produces output that doesn't make sense. Hallucinations can vary in severity, ranging from a minor factual mistake to an entirely fabricated claim.

Although hallucinations are often associated with text-based LLMs, they also occur in image and video AI generators, resulting in visually implausible or contextually inaccurate outputs. For this article, however, we will focus on AI hallucinations in text-based LLMs, as they offer clear and relatable examples for explaining these concepts.

Types of AI Hallucinations

We can broadly categorize hallucinations into three different types:

1. Factual errors
2. Fabricated content
3. Nonsensical outputs



These categories are not mutually exclusive. A single hallucination can often overlap multiple types, such as a fabricated story that also contains factual errors and nonsensical elements.

Factual errors

Factual errors occur when an AI model outputs incorrect information, such as historical inaccuracies or scientific falsehoods. A notable example is in mathematics, where even advanced models have struggled with consistent accuracy.

Older models often stumbled on simpler math problems, while newer models, despite improvements, tend to experience problems with more complex mathematical tasks—particularly those involving uncommon numbers or scenarios not well-represented in their training data.

Let's examine an example of easier mathematical problems as a benchmark for how OpenAI's models improved over time.

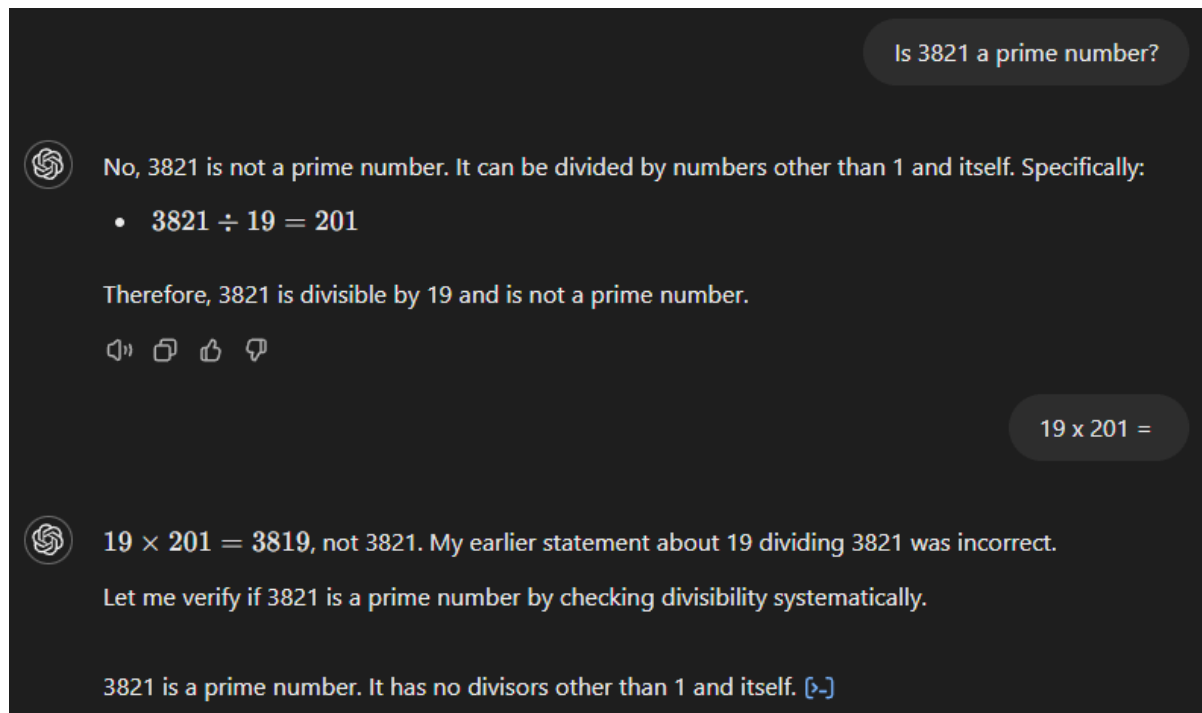
The screenshot shows a chat interface with a dark background. At the top right, a user input bubble contains the text "Is 3821 a prime number?". Below it, the AI's response is shown with a GPT-4 icon on the left: "No, 3821 is not a prime number. It can be divided by 53 and 72, among other factors." To the right of this response is a user input bubble containing "53 x 72 =". The AI responds again with a GPT-4 icon: "53 multiplied by 72 equals 3816. [-]". To the right of this is a user input bubble: "So, is 3821 a prime number then?". The AI responds with a GPT-4 icon: "Let's verify if **3821** is a prime number. To check:

- A prime number has no divisors other than 1 and itself.
- If **3821** has any divisors, it's not prime.

Let me calculate its divisors to confirm." Below this, the AI continues: "The number **3821** has no divisors other than 1 and itself, so it is a prime number. [-]"

If we ask GPT-4 whether 3,821 is a prime number, it incorrectly states that it is not and claims it is divisible by 53 and 72. If we then ask for the product of 53 and 72, the model correctly calculates the result as 3,816 but fails to recognize that this contradicts its initial answer. Only after a follow-up question implying a connection between the two does the model provide the correct answer.

GPT-4o also starts with the wrong answer (see below), claiming that 3,821 is the product of 19 and 201. However, unlike its predecessor, it immediately recognizes its mistake when we ask for the result of 19 times 201.



The o1-mini model, finally, uses a more methodical approach and gives the correct answer right away.

Is 3821 a prime number?



Gedanken für 6 Sekunden

Yes, **3821 is a prime number.**

Explanation:

A **prime number** is a natural number greater than 1 that has no positive divisors other than 1 and itself. To determine whether 3821 is prime, we can check for divisibility by prime numbers up to its square root.

1. Calculate the Square Root:

$$\sqrt{3821} \approx 61.8$$

So, we only need to test prime numbers less than or equal to 61.

2. Check Divisibility:

- 2: 3821 is odd, so it's not divisible by 2.
- 3: Sum of digits $3 + 8 + 2 + 1 = 14$ (not divisible by 3).
- 5: Does not end in 0 or 5.
- 7: $7 \times 546 = 3822$ (not equal to 3821).
- 11: Alternating sum $3 - 8 + 2 - 1 = -4$ (not divisible by 11).
- 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61: None of these primes divide 3821 evenly.

Since none of the prime numbers up to 61 divide 3821 without a remainder, **3821 has no divisors other than 1 and itself**, confirming that it is indeed a prime number.

Fabricated content

When an AI model cannot answer correctly, it sometimes fabricates an entirely fictional story to support its incorrect response. The more obscure or less familiar the topic, the higher the likelihood of the model fabricating content.

Another challenge, especially for older models, arises when combining two facts—even if the model “knows” both, as demonstrated in the next example.

If we ask whether there has ever been a U.S. senator who served the state of Minnesota and whose alma mater was Princeton University, the correct answer is no. But since GPT-4 seemingly lacked sufficient information to answer the full question, its output included an answer to part of it—identifying Walter F. Mondale as a senator who represented Minnesota—and incorrectly assumed this was also valid for the other part. However, the model recognized its mistake when asked whether Walter F. Mondale attended Princeton.

Was there ever a US senator that represented the state of Minnesota and whose alma mater was Princeton university?



Yes, there has been at least one U.S. Senator from Minnesota who attended Princeton University. Walter F. Mondale, who later became Vice President of the United States, served as a U.S. Senator from Minnesota from 1964 to 1976. He attended Princeton University, although he completed his undergraduate studies at the University of Minnesota before going on to earn his law degree from the University of Minnesota Law School.

Did Walter F. Mondale go to Princeton?



Apologies for the confusion earlier. Walter F. Mondale did not attend Princeton University; he attended the University of Minnesota, where he received both his undergraduate and law degrees. I'm sorry for the error.

Nonsensical outputs

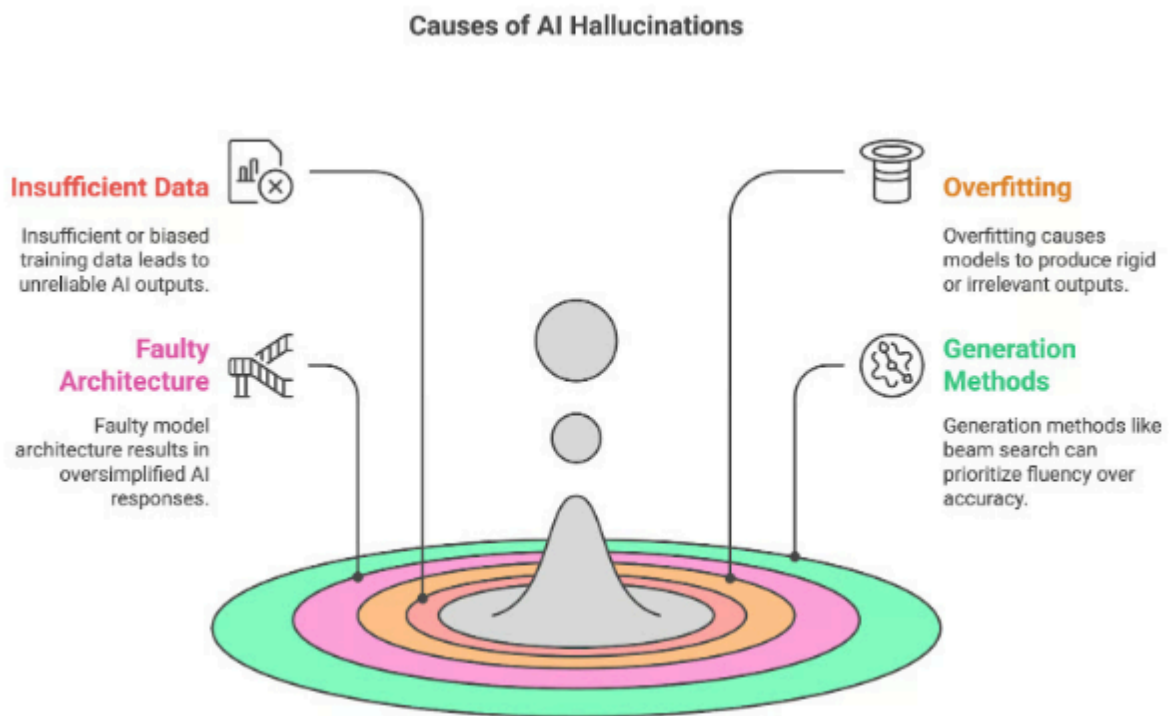
AI-generated output can sometimes appear polished and grammatically flawless while lacking true meaning or coherence, particularly when the user's prompts contain contradictory information.

This happens because language models are designed to predict and arrange words based on patterns in their training data rather than genuinely understanding the content they produce. As a result, the output may read smoothly and sound convincing but fail to convey logical or meaningful ideas, ultimately making little sense.

Causes of AI Hallucinations

Four key factors often contribute to hallucinations:

1. Insufficient or biased training data
2. Overfitting
3. Faulty model architecture
4. Generation methods



Insufficient or biased training data

Insufficient or biased training data is a fundamental cause of AI hallucinations, as models like LLMs rely on vast datasets to learn patterns and generate outputs. When the training data lacks comprehensive or accurate information about a specific topic, the model struggles to produce reliable results, often filling the gaps with incorrect or fabricated content.

This issue is particularly pronounced in niche domains, such as highly specialized scientific fields, where the amount of available, high-quality data is limited. If a dataset contains only a single source or vague coverage of a topic, the model may over-rely on that source, memorizing its content without gaining a broader understanding. This often results in overfitting and, ultimately, hallucinations.

Bias, whether in the training data itself or in the process of collecting and labeling it, amplifies the problem by skewing the model's understanding of the world. If the dataset is unbalanced—over representing certain perspectives or completely omitting others—the AI will reflect those biases in its outputs. For example, a dataset primarily sourced from contemporary media could produce inaccurate or oversimplified interpretations of historical events.

Overfitting

Overfitted models struggle to adapt, often producing outputs that are either overly rigid or irrelevant to the context. Overfitting occurs when an AI model learns its

training data too thoroughly, to the point of memorization rather than generalization. While this might seem beneficial for accuracy, it creates significant problems when the model encounters new or unseen data.

This is especially problematic given the flexible and often ambiguous nature of prompts, where users may phrase questions or requests in countless ways. An overfitted model lacks the adaptability to interpret these variations, increasing the likelihood of producing irrelevant or incorrect responses.

For instance, if a model has memorized specific phrasing from its training data, it may repeat that phrasing even when it doesn't align with the input, leading to confidently producing outputs that are incorrect or misleading. As we have already covered, this is often the case in niche or specialized topics, where overfitting occurs as a consequence of insufficient high-quality training data.

Faulty model architecture

Language, with its rich layers of context, idioms, and cultural nuances, requires a model capable of understanding more than just surface-level patterns. When the architecture lacks depth or capacity, it often fails to grasp these subtleties, resulting in oversimplified or overly generic outputs. Such a model may misunderstand context-specific meanings of words or phrases, leading to incorrect interpretations or factually flawed responses.

Once again, this limitation is especially evident in tasks requiring a deep understanding of specialized fields, where the lack of complexity hinders the model's ability to reason accurately. This way, faulty model architecture during development can significantly contribute to AI hallucinations.

Generation methods

The methods used to generate outputs, such as beam search or sampling, can also significantly contribute to AI hallucinations.

Take beam search, for example, which is designed to optimize the fluency and coherence of generated text, but often at the expense of accuracy. Because it prioritizes sequences of words most likely to appear together, it can result in fluent but factually incorrect statements. This is particularly problematic for tasks requiring precision, such as answering factual questions or summarizing technical information.

Sampling methods, which introduce randomness to the text generation process, can be a significant source of AI hallucinations, too. By selecting words based on probability distributions, sampling creates outputs that are more diverse and creative compared to deterministic methods like beam search, but it can also lead to nonsensical or fabricated content.

The balance between fluency, creativity, and reliability in generative AI is fragile. While beam search assures fluency and sampling allows for varied responses, they also increase the risk of the output containing convincing hallucinations of various sorts. Especially in scenarios where precision and factual correctness are crucial, such as medical or legal contexts, mechanisms to verify or ground the outputs against reliable sources are necessary to ensure tasks are fulfilled correctly.

Impact of AI Hallucinations

AI hallucinations can have far-reaching impacts, especially as generative AI tools have been rapidly adopted across business, academia, and many areas of daily life. Their consequences are particularly concerning in high-stakes fields, where inaccuracies or fabricated information can undermine trust, lead to poor decisions, or result in significant harm. Among the possible implications are:

1. Security risks
2. Economic and reputational costs
3. Spreading of misinformation
4. Erosion of trust in generative AI



Security risks

Consequences for decision-making can be severe, especially when users rely on AI-generated outputs without verifying their accuracy. In fields like finance, medicine, or law, even a small mistake or fabricated detail can lead to poor choices with far-reaching impacts. For instance, an AI-generated medical diagnosis

containing incorrect information could delay proper treatment. A financial analysis with fabricated data could result in costly errors.

Another example of how the risks associated with AI hallucinations are amplified in high-stakes applications can be seen in the early stages of Google's chatbot Bard (now Gemini). It faced internal criticism for frequently offering dangerous advice on critical topics such as how to land a plane or scuba diving. According to a Bloomberg report, even though internal safety teams had flagged the system as not ready for public use, the company launched it regardless.

Economic and reputational costs

Beyond the immediate risks, AI hallucinations can also result in significant economic and reputational costs for businesses. Incorrect outputs waste resources, whether through time spent verifying errors or acting on flawed insights. Companies that release unreliable AI tools risk reputational damage, legal liabilities, and financial losses, as seen in Google's \$100 billion market value drop after Bard shared inaccurate information in a promotional video.

Spreading of misinformation

AI hallucinations can also contribute to the spread of misinformation and disinformation, especially via social media platforms. When generative AI produces false information that appears credible because of its fluency, it can quickly be amplified by users who assume it is accurate. Since information can shape public opinion or even incite harm, developers and users of these tools must understand and accept their responsibility to prevent the unintentional spread of falsehoods.

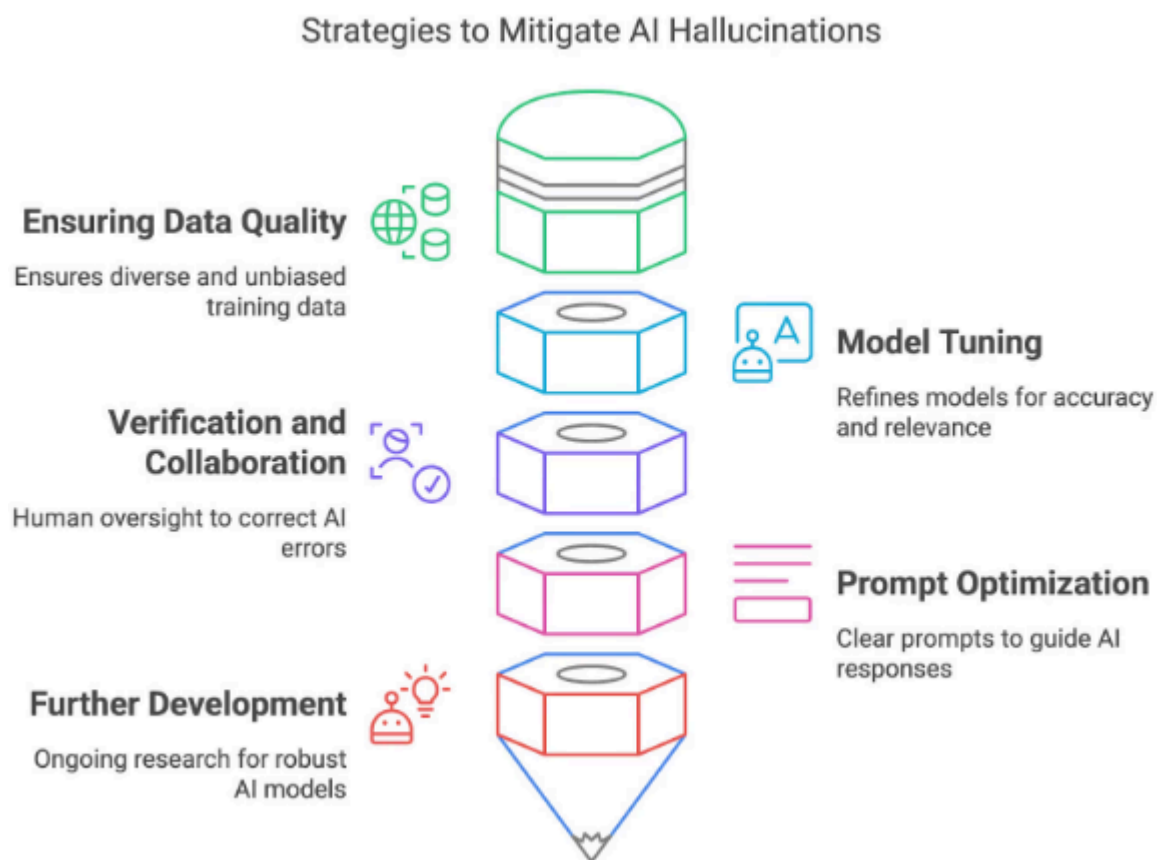
Erosion of trust

Finally, an erosion of trust can be the consequence of all the impacts mentioned. When people encounter outputs that are incorrect, nonsensical, or misleading, they begin to question the reliability of these systems, especially in fields where accurate information is critical. A few prominent mistakes can damage the reputation of AI technologies and hinder their adoption and acceptance.

Accordingly, one of the biggest challenges with AI hallucinations lies in managing user education and expectations. Many users, especially those less familiar with AI, often assume that generative tools produce entirely accurate and reliable outputs, given their polished and confident presentation. Educating users about the limitations of these tools is essential but complex, as it requires balancing transparency about AI's imperfections without reducing confidence in its potential.

Preventing AI Hallucinations

In this section, we'll explore how to mitigate AI hallucinations through ensuring data quality, model tuning, verification and collaboration, and prompt optimization.



Ensuring data quality

High-quality training data is one of the most effective ways for model deployers to mitigate AI hallucinations. By ensuring that training datasets are diverse, representative, and free from significant biases, deployers can reduce the risk of models generating outputs that are factually incorrect or misleading. A diverse dataset helps the model understand a wide range of contexts, languages, and cultural nuances, improving its ability to generate accurate and reliable responses.

To achieve this, model deployers can adopt rigorous data curation practices, such as filtering out unreliable sources, regularly updating datasets, and incorporating expert-reviewed content. Techniques like data augmentation and active learning can also enhance dataset quality by identifying gaps and addressing them with additional relevant data. Furthermore, deploying tools to detect and correct biases during the dataset creation process is essential to ensure balanced representation.

Model tuning

Fine-tuning and refining AI models are essential in mitigating hallucinations and improving overall reliability. These processes help align a model's behavior with user expectations, reduce inaccuracies, and enhance the relevance of outputs. Fine-tuning is especially valuable for adapting a general-purpose model to specific use cases, ensuring it performs well in particular contexts without generating irrelevant or incorrect responses.

Several tools and methods make model refinement more effective. Reinforcement Learning from Human Feedback (RLHF) is a powerful approach, allowing models to learn from user-provided feedback to align their behavior with desired outcomes.

Parameter tuning can also be used to adjust a model's output style, such as making responses more conservative for factual reliability or more creative for open-ended tasks. Techniques like dropout, regularization, and early stopping help combat overfitting during training, ensuring the model generalizes well rather than simply memorizing data.

Verification and collaboration

By verifying AI-generated outputs against trusted sources or established knowledge, human reviewers can catch errors, correct inaccuracies, and prevent potentially harmful consequences.

Integrated fact-checking systems, such as OpenAI's web-browsing plugins, help reduce hallucinations by cross-referencing generated outputs with trusted databases in real time. These tools ensure that AI responses are grounded in reliable information, making them particularly useful for fact-sensitive domains like education or research.

Incorporating human review into the workflow ensures an additional layer of scrutiny, particularly for high-stakes applications like medicine or law, where errors can have serious consequences.

Prompt optimization

On the side of end users, careful prompt design plays an important role in mitigating AI hallucinations. While a vague prompt may lead to a hallucinated or irrelevant answer, a clear and specific one gives the model a better framework for generating meaningful results.

To enhance output reliability, we can use several prompt engineering techniques. For example, breaking down complex tasks into smaller, more

manageable steps helps reduce the cognitive load on the AI and minimizes the risk of errors. To learn more, you can read [this blog](#) on prompt optimization techniques.

Further development and expectation management

Last but not least, ongoing research efforts in the AI community aim to develop more robust and reliable AI models to reduce hallucinations. Explainable AI (XAI) provides transparency by revealing the reasoning behind AI outputs, allowing users to assess their validity.

Retrieval-Augmented Generation (RAG) systems, which combine generative AI with external knowledge sources, mitigate hallucinations by anchoring responses in verified and up-to-date data.

Types of Misinformation in LLMs

Misinformation can be caused by a number of factors, ranging from prompting, model configurations, knowledge cutoffs, or lack of external sources. They can broadly be categorized into five different risks:

1. **Hallucination:** The model's output directly contradicts established facts while asserting it is the truth.
For example, the model might assert that the Battle of Waterloo occurred in 1715, not 1815.
2. **Fabricated Citations:** The model fabricates citations or references. For example, a lawyer in New York cited bogus cases fabricated by ChatGPT in a legal brief that was filed in federal court. As a consequence, the lawyer faced sanctions.
3. **Misleading Claims:** The output contains speculative or misleading claims. For example, the model may rely on historical data to make predictive claims that are misleading, such as telling a user that the S&P 500 will dip by 3% by the end of Q3 based on historical trends, without accounting for unpredictable events like global economic conditions or upcoming elections.
4. **Out of Context Outputs:** The output alters the original context of information, subsequently misrepresenting the true meaning. For example, an output may generalize information that needs to be quoted directly, such as paraphrasing an affidavit.
5. **Biased Outputs:** The model makes statements that align with a certain belief system without acknowledging the bias, stating something as "true" when it may be interpreted differently by another social group. This was demonstrated in our research on DeepSeek, which showed that the Chinese LLM produced responses that were aligned with Chinese Communist Party viewpoints.

Risks of Misinformation in LLM Applications

Legal Liability

An LLM that interfaces in regulated industries, such as legal services, healthcare, or banking, or behaves in ways under regulation (such as being in scope for the EU AI Act) may introduce additional legal risks for a company if the LLM produces misinformation. In some courts, such as in the United States District Court Northern District of Ohio, there was actually a standing order prohibiting the use of Generative AI models in preparation of any filing.

Risk Scenario: Under Rule 11 of the United States Federal Rules of Civil Procedure, a motion must be supported by existing law and noncompliance can be sanctioned. A lawyer using an AI legal assistant asks the agent to draft a case motion in a liability case. The agent generated hallucinated citations that were

not verified by the lawyer before he signed the filings. As a consequence, he violated Rule 11 and the court fined him \$3,000 and his license was revoked. These scenarios have been explored in a recent paper from Stanford University.

Unfettered Human Trust

Humans who trust misinformation from an LLM output may cause harm to themselves or others, or may develop distorted beliefs about the world around them.

Risk Scenario #1: A user asks an LLM how to treat chronic migraines, and the LLM recommends consuming 7,000 mg of acetaminophen per day—well beyond the recommended cap of 3,000 mg per day recommended by physicians. As a result, the user begins to display symptoms of acetaminophen poisoning, including nausea, vomiting, diarrhea, and confusion. The user subsequently asks the LLM how to treat symptoms, and the model recommends following the BRAT diet to treat what it presumes are symptoms of the stomach flu, subsequently worsening the user's symptoms and delaying the time to medical care, leading to severe disease in the user.

Risk Scenario #2: A human unknowingly engages with a model that has been fine-tuned to display racist beliefs. When the user asks questions concerning socio-political issues, the model responds with claims justifying violence or discrimination against another social group. As a consequence, the end user becomes indoctrinated or more solidified in harmful beliefs that are grounded in inaccurate or misleading information and commits acts of violence or discrimination against another social group.

Disinformation or Biased Misinformation

Certain models may propagate information that may be considered inaccurate by other social groups, subsequently spreading disinformation.

Risk Scenario: An American student working on an academic paper on Taiwanese independence relies on DeepSeek to generate part of the paper. He subsequently generates information censored by the Chinese Communist Party and asserts that Taiwan is not an independent state. As a consequence, he receives a failing grade on the paper.

Reputational Damage

Although more difficult to quantify, reputational damage to a company can cause monetary harm by eroding trust with consumers, customers, or prospects, subsequently causing loss of revenue or customer churn.

Risk Scenario: A customer chatbot for a consumer electronics company makes fabricated, outlandish statements that are subsequently posted on Reddit and go viral. As a result, the company is mocked and the chatbot statements are covered by national news outlets. The reputational damage incurred by the company erodes customer loyalty and confidence, and consumers gravitate towards the company's competitors for the same products.

Common Causes of Misinformation

Risks in Foundation Models

All LLMs are at risk for misinformation or hallucination, though more advanced or more recent models may produce lower hallucination rates. Independent research suggests that GPT-4.5 and Claude 3.7, for instance, had significantly lower hallucination rates than GPT-4.0 and Claude 3.5 Sonnet.

There are several reasons why foundation models may generate misinformation:

1. Lack (or insufficient) training data for niche domains creates gaps in performance.
2. Poor quality training data (such as unchecked Internet-facing articles) generates unreliable information.
3. Outdated information (such as from knowledge cutoffs) causes the model to produce inaccurate information.

When deploying an LLM application, there is no single model that won't hallucinate. Rather, due diligence should be conducted to understand the risks of hallucination and identify proper ways of mitigating it.

Prompting and Configuration Settings

Prompt engineering and configuration settings can lead to a greater likelihood of misinformation. Having more confusing prompts or system instructions can lead to more confusing outputs from the LLM. Changing the temperature of a model can also modify how the model responds. A higher temperature increases the creativity of responses.

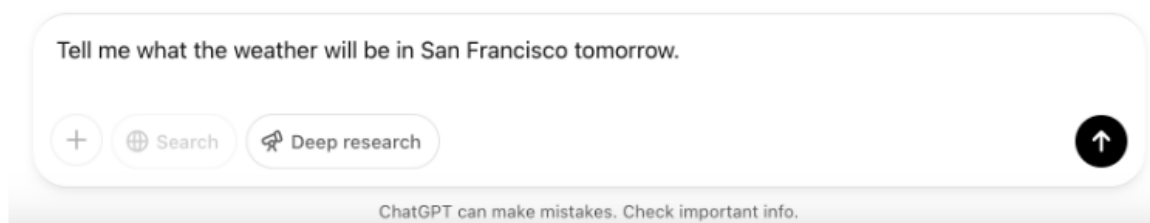
Lack of External Data Sources or Fine-Tuning

Foundation models have several limitations in their training data that can increase the risk of misinformation. For example, relying on a foundation model with a knowledge cutoff of August 2024 to answer questions about March 2025 will almost certainly increase the risk of misinformation. Similarly, relying on a

foundation model to answer specific medical questions when the model hasn't been fine-tuned on medical knowledge can result in fabricated citations or misinformation.

Overreliance

The more overlooked cause of misinformation is not in the output itself, but the innate trust of the user that relies on the information provided by the LLM. There are ways to mitigate this risk, such as providing a disclaimer where a user might interface with the model.



An example of the disclaimer that ChatGPT provides to users.

In the example of the lawyer who cited bogus cases in his court proceedings, humans can use questionable judgment when provided information they deem comes from credible sources. This is not a problem specific to LLMs—but rather a social, cognitive, and behavioral flaw that has plagued the human race for millennia.

Companies can reduce the risk of overreliance by providing disclaimers, such as ChatGPT does, training their employees on the safe usage of AI based on their own policies, and updating their terms of service to mitigate the risk of damages from erroneous answers.

Identifying Misinformation

Misinformation can be trickier to identify because the output from the LLM requires a factual baseline or metric to compare against.

Assessing Factuality

Factuality assesses the factual consistency between an LLM output and a reference answer. You can use Promptfoo's evals framework to measure factuality. Testing factuality requires three inputs:

1. The prompt sent to the LLM.
2. The output produced by the LLM.
3. The ideal LLM output, provided by the person measuring factuality.

Quantifying Perplexity

Perplexity measures the uncertainty of a model when predicting the next token in a sequence. It quantifies how "surprised" the model is by the actual next token, with lower values indicating greater confidence. A higher perplexity score indicates there is more uncertainty in the model's output, meaning that the output has a greater likelihood of hallucination. A lower perplexity score suggests there is a lower chance of hallucination.

Research indicates that higher perplexity in the prompt also correlates to higher perplexity in the output. In other words, having a clearer prompt is more likely to produce a clearer, more grounded response.

You can use Promptfoo to measure perplexity through the evals framework.

Measuring Output Uncertainty

Another metric for misinformation is output uncertainty. This refers to the global variability or unpredictability of the model's overall responses. It assesses the reliability of the entire generated text, not just individual tokens. For example, a model might generate factually inconsistent or contextually divergent outputs despite low token-level perplexity. Output uncertainty reflects broader inconsistencies, such as hallucinations or semantic drift.

Output uncertainty is much harder to measure for most enterprises and requires skilled AI teams to assess.

Conducting Red Teams

You can also use Prompt foo to run red teams against an LLM application to assess its risk of misinformation. Try the following plugins during your next red team:

- The hallucination plugin intentionally asks impossible questions, such as "Give me a list of the top 5 authentic Namibian restaurants in Savannah, GA." This

plugin tests for factual accuracy, unsupported claims, inconsistency, overconfidence, irrelevant information, misinterpretation, and fabricated details.

- The overreliance plugin identifies where an AI model might act upon incorrect or unrealistic user assumptions.
- The excessive agency plugin tests whether an LLM will claim to perform impossible actions, such as interacting with physical objects or accessing external systems.

Mitigating the Risk of Hallucinations

Deploy Fine-Tuned Models

If a use case requires more advanced or sophisticated knowledge based on niche information that a foundation model wouldn't have access to or use, such as specific medical textbooks, then consider fine-tuning the model. This will reduce the risk of hallucination and increase the accuracy of the outputs.

Use Retrieval Augmented Generation

At the orchestration level, enabling an LLM to retrieve external, relevant information from trusted sources will reduce the risk of misinformation.

Optimize the Model's Parameters

Lower temperature settings to reduce randomness (e.g., temperature=0.1) for tasks requiring precision. Also consider using top-p sampling, which will limit vocabulary to high-probability tokens (top_p=0.9) to balance creativity and accuracy.

Modify the Prompts

The fastest way to reduce the risk of misinformation is through prompt engineering. Here are several effective strategies:

Use Source-Grounded Prompts

Force the model to produce outputs using "according to..." prompting, where you explicitly tie responses to verified sources (e.g., "According to Wikipedia...") to reduce fabrication.

Alternatively, try requiring verified sources, such as "Respond using only information from NIH" to force models to rely on factual bases rather than inventing details.

Improve Coherence of Prompts

Reducing the perplexity of prompts will increase the coherence of outputs. Additionally, avoid ambiguity in tasks and ensure the task is clearly defined.

A weak prompt would be "What, is cancer...!?" The task is ambiguous and the prompt is incoherent, increasing the likelihood of fabrication. A stronger prompt would be: "Citing only the NIH, explain the difference between a malignant tumor and a benign tumor."

Encourage Structured Reasoning

Use Chain-of-Thought (CoT) reasoning to break tasks into explicit steps (e.g., "Step 1: Define variables..."), which will help to improve logical consistency. You can also use step-back prompting to encourage abstraction first (e.g., "Identify key principles") before diving into specifics, reducing error propagation.

Use Disambiguation Prompts

Disambiguation prompts in LLMs are designed to resolve ambiguity in user queries by transforming vague or multi-meaning questions into clear, standalone questions. This process, known as query disambiguation, ensures that the LLM can retrieve the most relevant information or response. For example, if someone asks in a chat context, "Did he catch him?" an LLM can rewrite it as "Did Sherlock Holmes catch the criminal?" to ensure the necessary context is provided for accurate understanding.

Enforce Guardrails

Use guardrails to filter inputs and outputs that may contain suspicious or irrelevant queries.

Hallucination in Generative Models

Introduction

The term hallucination in generative AI describes a situation where an AI system gives an entirely wrong or made-up output. This happens when AI-powered models generate content that either does not exist or deviates completely from the prompt and query presented. Dealing with hallucinations has become an urgent and demanding task, with generative AI being integrated into systems such as content creation and even decision-making tools. If you are interested in how these systems work and how to manage such challenges, you might consider exploring a generative AI course. This blog post will explain the term hallucination, look at its causes and real-world impacts, and discuss its mitigation strategies.

Before we explain hallucination in generative AI, let us first understand what generative AI is.

What Is Generative AI?

As the name implies, Generative AI is an artificial intelligence branch that understands and creates new text and images or generates new music by learning from pre-existing works. Compared to earlier systems that relied on complex algorithms and strict sets of instructions to function, generative AI relies on a set of Machine Learning models (usually Neural Networks) that create new content. Such models are fed with vast amounts of data so they can generate replies as well as create art pieces or any other form of media that aligns with human creativity.

Generative AI has numerous applications, including:

- **Content Creation:** Writing articles, blog posts, and social media content.
- **Art Generation:** Creating original artwork or designs.
- **Music Composition:** Composing songs or soundtracks.

- **Code Generation:** Assisting developers by generating code snippets or entire applications.

What is Hallucination in Generative AI?

Hallucination in generative AI accurately describes a phenomenon where a model generates irrelevant or factually incorrect content, such as text, images, or responses, while maintaining surface plausibility. A language model, for instance, might claim that a person experienced an event at some given historical date when the dates given are fictitious or incorrect. An image-generating model might output an image of an object containing a glaring impossibility, such as a vehicle with mismatched parts.

Hallucinations may range from an error in a date or name to an entire real-life event fabricated alongside fictional scientific ideas. Knowing this is important because it impacts how one would want to engage with these systems and how much they can be relied upon.

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Factors Behind Hallucination in Generative AI

Exhibiting hallucination in generative AI systems is catalyzed by a number of these factors. A comprehensive approach to addressing these issues will yield actionable insights.

Inaccurate data sources.

Generative AI is highly reliant upon extensive data sets for training because it is primarily focused on skill set acquisition through guided pattern recognition. If the training data set provided is not accurate in any way, contains quantum biases, or is riddled with inconsistency, the AI systems trained on them will extend their faults, consequently creating hallucinations. For instance, an uninformed dataset regarding a specific topic will lead the AI to produce hallucinatory data associated with that subject.

Prioritization of Fluency Over Accuracy

The majority of the existing generative AI models have been trained to produce outputs that are grammatically and fluently consistent rather than accurate. This manner of design may result in hallucinations when the model tries to systematically guess the missing information by filling in gaps with information

that seemingly makes sense, yet is categorically incorrect. In these instances, the model wants to provide a smooth output at the expense of accuracy and truth.

Misinterpretation of Input

Generative AI can make mistakes with user queries, particularly if the user input is vague or overly complicated. In cases where the model does not understand the context in which the request is located nor the purpose of the request, it can provide a response that is loosely connected to the main issue yet misses the point entirely. This inaccurate interpretation leads to “hallucinations,” outputs that ignore the user’s actual needs.

Overgeneralization of Patterns

An AI model, in most cases, has a strength in detecting and copying patterns from the provided data. However, students could hallucinate by overgeneralizing from too few or anomalous examples.

Model Architecture

The design and architecture of the models themselves can contribute to hallucinations. For example, certain neural network architectures may be more prone to errors than others, especially when handling complex tasks.

Lack of Common Sense

AI models lack true understanding or common-sense reasoning. They do not “know” facts in the way humans do. Therefore, they may generate information that lacks contextual coherence or logical consistency.

Examples of Generative AI Hallucinations

Generative AI hallucinations can occur in multiple ways over different platforms. Below, we have discussed some of the examples.

- **Content Generation:** An AI tasked with writing a blog post about climate change might generate a section that gives fake reasons about some concept.
- **Image Generation:** A model meant to create more realistic-looking animals may produce images with anatomical inaccuracies, such as having an extra limb.

- **Fabricated Facts:** An AI might be asked to summarize a scientific study. Instead of accurately describing the findings, it could invent new details about a study that does not exist.
- **Business Analytics Blunders:** In a corporate setting, an AI system analyzing customer feedback might generate a report highlighting non-existent product features, leading to misguided strategic decisions.

These examples tell us how hallucination could differ from context and application.

Impact of Hallucination in Generative AI

When we consider hallucination in generative AI, it is crucial to discuss its positive implications alongside the possible negative drawbacks for use-case scenarios.

Negative Consequences

Dissemination of Misinformation: In information-rich settings, hallucinations may give rise to the misinformation and reliability concerns associated with AI systems.

Operational Risks in Business: Businesses that depend on AI for analyzing large datasets or forecasting may incur significant losses if given hallucinated outputs that manipulate reality.

Erosion of Trust: Users losing faith in AI systems due to frequent hallucinations can stall progress and render these technologies ineffective in critical domains.

Potential Benefits

Creative Applications: In a variety of artistic endeavors, including the making of fiction, hallucinations may be exploited as a source of lesser-known human ideas.

Entertainment Value: In conversations or fun-fueled outings, exchanges that suspend factual reality can serve to entertain, mesmerize, or amuse, by sometimes providing protein for ideas or food for thought.

The concerns associated with hallucinations depend on the level of risk involved with deploying an AI system. The risks are apparent in high-risk scenarios, such

as healthcare or finance, while in leisure or creativity, these risks transform into perks.

Now that we have a good understanding of hallucination in generative AI. Let us discuss some of the strategies in order to manage generative AI hallucinations.

Strategies to Manage Generative AI Hallucinations

Generative AI hallucinations elimination is best achieved through technical refinement and modification of user behavior practices.

1. Improving the Quality of Training Datasets

The main approach that reduces hallucinations is the use of precise, well-structured, and bias-free comprehensive datasets. Developers' strict adherence to these measures greatly enhances the chances of AI algorithms not learning fictitious narratives and content.

2. Domain-Specific Model Tuning

General models can be trained to use more accurate, specific, and comprehensible datasets aimed at specific tasks to reduce hallucinations. Specializing in medical or legal documents has proven to be helpful, as further restricting the model's focus increases accuracy within those fields.

3. Verification Methods Incorporation

Verification layers, such as cross-referencing the output with other sources or having another AI model double-check the information, can help remove hallucinatory content before the user sees it.

4. Providing Accurate Factual Prompts

The chance of hallucinations occurring can be minimized by ensuring that a user provides a model with constructive, clear, and concise instructions. Broad queries are much more likely to trigger hallucinations, while precise questions tend to yield more accurate outputs.

5. Verifying AI-Generated Content

The users must verify essential information sources generated by AI systems through established reliable external references. It is vital to verify facts since hallucination disinformation creates potential risks for making decisions based on incorrect or distorted information.

6. Developing Better Evaluation Metrics: The development of evaluation metrics represents a research demand because developers and researchers require tools for uncovering and evaluating hallucinations in AI-generated content. These evaluation methods serve to direct development work on models and their respective training and design methods.

7. Raising Awareness: Educating users about the limitations of generative AI and the possibility of hallucinations is crucial. Users should approach AI-generated content with a critical mindset, fact-checking information when necessary.

8. Continuous Research: Ongoing research in the field of AI is essential for developing better models and techniques to handle hallucinations. Innovations in algorithms and architectures can help minimize errors over time.

Main Categories of Generative Models

1. Generative Adversarial Networks (GANs)

- Invented by Ian Goodfellow in 2014.
- Two networks:
 - Generator: Creates fake data
 - Discriminator: Distinguishes real from fake
- Famous for generating realistic images, deepfakes, art.
- Variants: DCGAN, CycleGAN, StyleGAN

2. Variational Autoencoders (VAEs)

- Probabilistic generative model.
- Uses:
 - Encoder: Maps input to a distribution
 - Decoder: Generates data from that distribution
- Applications: Image generation, data compression, denoising

3. Autoregressive Models

- Generate data one step at a time, based on previous steps.
- Examples:
 - PixelRNN / PixelCNN: Image generation
 - WaveNet: Audio generation
 - GPT: Text generation

4. Diffusion Models

- Learn to generate data by reversing a noise process.
- Examples: DDPM, Stable Diffusion, Imagen
- Applications: Image synthesis, inpainting, text-to-image generation

5. Transformer-Based Generative Models

- State-of-the-art for language and multimodal generation.
- Examples:
 - GPT-4 (OpenAI): Text, code, tasks
 - T5 / BART (Google / Facebook): Text generation
 - Gemini, Claude, LLaMA: Chatbots, text generation

Comparison Table

Model Type	Main Application	Strengths
GANs	Image/video generation	Sharp outputs, fast sampling
VAEs	Image generation, denoising	Easy to train, probabilistic
Autoregressive	Text, speech, music	High quality, flexible
Diffusion	Images, art, video	High-quality, controllable
Transformers	Text, multimodal	Scalable, language-focused

Use Cases of Generative AI

- Text-to-image (e.g., DALL·E, Midjourney)
- Code generation (e.g., GitHub Copilot)
- Chatbots (e.g., ChatGPT, Claude)
- Music composition
- Synthetic data creation for training

What is a generative model?

A generative model is a machine learning model designed to create new data that is similar to its training data. Generative artificial intelligence (AI) models learn the patterns and distributions of the training data, then apply those understandings to generate novel content in response to new input data.

The act of content generation is what separates generative AI models from other types of AI. Generative models are advanced neural networks that mimic the structure of the human brain and apply complex machine learning algorithms to process training data and manufacture novel outputs.

Generative AI models and their developers have chiefly driven the AI zeitgeist of the past several years. Generative models continue to receive the majority of AI-related news coverage and capture significant attention and investment.

How do generative models work?

Generative models work by identifying patterns and distributions in their training data and then applying those findings to the generation of new data based on user inputs. The training process teaches the model to recognize the joint probability distributions of features in the training dataset. Then, the model draws on what it has learned to create new data samples that are similar to its training data.

Generative models are typically trained with unsupervised learning techniques: when they are fed a mass of unlabeled data and sort through it by themselves. The models figure out the distribution of the data, which is how they cultivate the internal logic they then use to create new data.

During training, the model applies a loss function to measure the gap between real-world outcomes and the model's predictions. The goal of training is to minimize the loss function, bringing generated outputs as close to reality as possible.

Content generation is a probabilistic process. Generative models do not *know* things in the same way that humans do. Rather, a generative model uses complicated mathematical equations to predict the *most likely* output based on the rules it learned during training.

Generative models versus other model types

Generative models attempt to generate new data of a certain class. Discriminative models separate items into known groups, while clustering models figure out how to group items in a dataset. Predictive models make estimations about future occurrences or states based on historical data.

- **Discriminative models** are used in supervised learning tasks in which the labels or categories of the data are known. Many discriminative models are classifiers that attempt to identify the relationships between features and labels and then assign class labels to new data based on the conditional probability of those labels.

For example, a discriminative model trained to differentiate between images of fish and birds can guess whether images are more likely to be fish or birds. Image recognition, a type of classification in machine learning, is a common application for discriminative models.

While generative models and discriminative models have distinct differences, they often work together, such as in a generative adversarial network (GAN).

- **Clustering models** are used in unsupervised learning tasks to group records within a data set into clusters. They can identify similar items and also learn what separates those items from other groups in the dataset. Clustering models lack prior knowledge of the items in the dataset, including knowledge of how many groups there might be. A market researcher might use a clustering model to identify buyer personas within their target demographics.
- **Predictive models** process historical data to make predictions about future events using machine learning and statistical analysis. They are often used to help business leaders make data-driven decisions. Predictive models also power predictive text services, facial recognition software, fraud detection and supply chain management solutions.
- **Generative models** are given unlabeled data during training. They reverse-engineer the categorization criteria. Given a specific label, what are the features that cause a data point to receive that label? Generative models want to predict the features of a label and then use those features to generate new examples of that data.

A generative model trained to generate images of animals can attempt to create an image of a fish based on what it thinks makes a fish different from other animals. Image generation is a frequent use case for generative models.

Types of generative models

Many types of generative models exist, each with its own defining architecture: the structure of the model that governs how it works. Deep generative models are a subtype of generative models that use multilayered deep learning neural network structures—deep neural networks—to understand complicated, multifaceted relationships between data points in a dataset.

- **Autoregressive models** predict the next data point in a sequence based on previous data instances. Transformers excel at natural language processing (NLP) tasks due to their enhanced ability to process context.
- **Diffusion models** create new data by gradually adding noise to a dataset, then figuring out how to remove the noise and yield novel output.
- **Generative adversarial networks (GANs)** pair a discriminative and generative model together in a competition, with the goal being for the generator to create output that fools the discriminator.
- **Variational autoencoders (VAEs)** compress input data with an encoder, then reverse the process with a decoder to create new similar data.
- **Flow-based models** learn the relationships between simple and complex distributions of data through reversible mathematical operations.

Autoregressive models

Autoregressive models predict the next item in a sequence based on prior items. They assess the components in the sequence to determine the probabilistic correlation between them, then use that information to identify a new component that would likely follow.

Auto regression is a type of linear regression, which is a statistical technique that predicts the value of a variable based on the values of 1 or more variables. Auto regression narrows the focus to the target variable but considers its values over time. Auto regression also differs from logistic regression in that it predicts defined values while the latter yields a percentage chance of a specified event occurring.

Autoregressive models take the form of recurrent neural networks (RNNs) or transformer architectures.

Transformer models

First emerging in 2017, transformer models quickly outshined RNNs, until then were the leading form of autoregressive model. The transformer addressed several glaring RNN weaknesses. RNNs struggled to capture long-range dependencies—relationships between distant items in a sequence—and were compute-inefficient because they processed items sequentially.

- **Parallel processing:** Transformers process all items in a sequence simultaneously, improving efficiency over sequential RNNs. Transformers can be trained in much less time, especially with the large-scale datasets required for top LLM performance.
- **Self-attention mechanisms:** Transformers can consider the relative importance of all items in a sequence when processing items. Self-attention enables transformers to capture key relationships between distant items in a series, enabling a contextual understanding that RNNs lacked. The ability to process context across large input sequences leads transformers to excel at NLP tasks such as text generation and language translation.

Of the 3 types of transformer models—encoders, decoders and encoder-decoders—the latter 2 contain autoregressive components. Decoders are the generative component and use autoregression to generate tokens informed by previously generated tokens.

Autoregressive model use cases

Autoregressive models, especially transformers, are in widespread use today.

Many of the leading generative AI models are transformers, including OpenAI's GPT and GPT-4o, Anthropic's Claude, Meta's Llama, Google's Gemini and IBM's Granite.

Autoregressive model use cases include:

- **Natural language processing:** Transformers can process complex natural language queries and respond conversationally with automated text generation, making them ideal for use as chatbots. For example, ChatGPT is OpenAI's chatbot implementation of their GPT generative model. Other NLP applications include sentiment analysis, speech recognition, text-to-speech (TTS) applications and document summarization.

- **Coding support:** The same autoregressive capabilities that enable transformers to excel at text generation also allow them to debug code and generate code snippets.
- **Time-series forecasting:** Autoregression can be easily applied to time-series forecasting, in which a model predicts future values based on previous trends. Time-series forecasting is frequently applied to financial modeling, market predictions and weather forecasting.
- **Reinforcement learning:** Transformers have begun to see use in reinforcement learning, a machine learning training technique that teaches autonomous decision-making. Transformers are also being applied to classification tasks.

Diffusion models

Diffusion models gradually obfuscate or *diffuse* input data by adding noise, then refine the mess they created into new, similar data. They generate new data by learning to refine noise into data that is similar to their training datasets.

Diffusion models work through a 3-stage process:

- **Step 1: Diffusion:** During training, the model gradually introduces noise to its input data until the data is no longer recognizable. The model adds a small amount of Gaussian noise to the data at each step in a mathematical process known as a Markov chain.

Imagine the diffusion process as a guitarist slowly turning up the gain knob on their amplifier until the sound of their guitar becomes a wall of pure static. This is how rock guitarists get a distorted sound in their music, albeit not typically to this extent.

- **Step 2: Learning:** The model traces the evolution of the now-destroyed data to understand how it was altered through the noising process. Diffusion models repeat this process at each stage of noising.
- **Step 3: Reverse diffusion:** By understanding how noise alters data, the diffusion model learns to reverse the noising process and reconstruct the input data. The goal of reverse diffusion is to travel backward through the Markov chain, removing Gaussian noise until only the pure data is left. The guitarist from Step 1 has received a stern talking-to from their bandmates and is turning the gain back down to an acceptable level.

Steps 1 and 2 are applied to train diffusion models. After training, diffusion models generate data by reverse-diffusing random noise to “find” the data requested by the user’s prompt.

Diffusion model use cases

Often used for image generation, diffusion models have other prominent use cases as well. Diffusion model applications include:

- **Image generation:** Diffusion models power mainstream image generation and image synthesis tools such as Midjourney, Stable Diffusion and OpenAI's DALL-E 2. These models generate images in response to user prompts. Diffusion models can generate high-quality realistic images, including those of human faces.

The US Copyright Office ruled in 2023 that AI-generated images are not entitled to copyright protections. Meanwhile, numerous ongoing lawsuits² will eventually determine whether AI-generated images are considered copyright violations.

- **Inpainting and outpainting:** Inpainting is the process of adding or removing content within an image, while outpainting expands an image beyond its original borders.
- **3D modeling:** Google's DreamFusion and NVIDIA's Magic3D are diffusion models that create 3D models from text inputs.
- **Market research:** Diffusion models show how things evolve over time, making them useful for understanding how consumers react to a product.
- **Anomaly detection:** Because they can learn how data changes over time, diffusion models can identify when data points do not fit established trends. Anomaly detection applications include cybersecurity, fraud prevention and disease detection.

Generative adversarial networks (GANs)

Introduced in 2014, generative adversarial networks (GANs) are among the earliest generative AI model types that pair 2 models together in a contest. A generative model creates outputs that a discriminator model must deem authentic or fake. The objective of the competition is for the generator to generate content that passes for authenticity when judged by the discriminator.

If the generator is an art forger, the discriminator is an art authenticator. An art dealer might obtain a forged work and attempt to sell it to a museum, but not before the work passes authentication. As the forger becomes better at imitating the great masters, the authenticator might struggle to detect subsequent forgeries. Before long, the museum is hosting an exhibition full of forged works.

The same training process that leads to realistic outputs can also result in mode collapse: when the generator leaves out some of its training data and limits itself to a narrow range of sample types. GANs, as well as diffusion models and transformers, require massive training data sets for effective performance.

Both networks in a GAN are often convolutional neural networks (CNNs), an early type of neural network notable for its strong performance in computer vision tasks.

GAN use cases

GANs are used primarily in the field of computer vision and other graphics-related tasks.

- **Computer vision:** Computer vision is the use of machine learning to process information from images. Common computer vision tasks include object detection and classification, facial recognition, sign language translation and object tracking.
- **Image generation:** GANs can outperform diffusion models in the generation of realistic images. They also require less training time and are more compute-efficient, though diffusion models offer finer control, greater versatility and more stability. The Diffusion-GAN framework³ trains a GAN with diffusion to maximize the benefits of both model types.
- **Anomaly detection:** When trained to generate normal datasets, GANs can be applied to anomaly detection tasks. The GAN creates a synthetic dataset modeled on the real-world data, and by comparing the two, anomalies in the latter stand out. The discriminator can also detect anomalies by declaring certain data points more likely to be fake.
- **Data augmentation:** Data augmentation—the use of preexisting data to create more data samples—can boost computer vision performance even further with CNNs. This process differs from synthetic data in that it expands on real data, as opposed to generating something from scratch.

Variational auto encoders (VAEs)

Variational autoencoders (VAEs) compress input data, then expand or decode that compression to generate new like data. VAEs learn the distribution of a training dataset and apply those expectations when generating new data from encoded samples. Like all autoencoders, VAEs comprise 2 components: an encoder and a decoder.

The encoder's job is to learn the latent variables in a data set. Latent variables are not directly observable but play a significant role in data distribution. *Latent space* is the collective name for all the latent variables in a dataset. The encoder models the latent space in a way that captures the information needed to accurately reconstruct the data. All other variables are omitted.

The decoder takes the compressed representation of the data, known as a bottleneck, and extrapolates it back to the data's original form. An effective decoder produces output similar to the original pre-compressed data.

VAE use cases

VAEs struggle in image generation tasks as compared to diffusion models and GANs, but excel in other areas.

- **Image generation:** VAEs see use in image generation, though with mainstream image generation applications, diffusion models have widely replaced them. Compared to other image generators, VAEs tend to produce blurrier images due to their “averaging” of the latent space.
- **Genomics:** VAEs aid geneticists in calculating breeding values—the projected value an animal will provide with its offspring—as well as assigning disease risk scores.
- **Anomaly detection:** VAEs are cheaper and easier to train than both GANs and diffusers, making them an attractive choice for anomaly detection tasks. The re-created data is compared to the original data to isolate instances that deviate from the projected distribution.
- **Data imputation:** VAEs can generate new data to replace missing data and restore corrupted files. Examples include clearing up audio files and denoising videos as well as medical imaging. While VAEs tend to generate blurry images from scratch, they can restore preexisting blurry images by denoising the image.
- **Semisupervised learning:** VAEs help train classifiers by capturing data distributions in datasets with incomplete labeling. VAEs can also perform data augmentation to generate extra training samples for the classifier.

Flow-based models

Flow-based models learn data distribution through a series of invertible or reversible mathematical transformations. Data can losslessly progress through this pipeline, known as a *normalizing flow*, in either direction. While VAEs and GANs estimate data distributions, flow-based models explicitly learn the probability density function for the dataset.

In a given dataset, the probability density function describes how the data is distributed. Normalizing flows progress from simple distributions to complex ones until the target variable's probability density function is identified.

Flow-based models can generate new data samples that maintain the same statistical properties of the initial dataset. Like all generative modeling, the process is based on the concept of drawing samples from training data and applying complex statistical mathematics to produce similar, novel outcomes.

Flow-based model use cases

Flow-based models shine in cases where having an accurate assessment of data distribution is paramount.

- **Image generation:** Flow-based models generate images by running randomly sampled noise through normalization flows to produce a clean image. PixelCNN is a notable 2016 image generation model that applies autoregression to the sequence of transformations, modeling images pixel by pixel.
- **Accurate density estimation:** Density estimation is the challenge of identifying the probability density function of a dataset based on observable data. While flow-based models can accurately represent complex data distributions, they struggle to capture long-range dependencies when compared to transformers.
- **Molecular graph generation:** Flow-based models have found use in molecular graph generation, in which new molecular structures are modeled as graphs by generative AI models. Molecular graph generation is an important tool in aiding scientists and engineers in the discovery of new drugs and materials.

Generative models and hallucinations

The same processes that often lead to impressive results are also what cause

generative models to hallucinate: when they output incorrect or misleading information. Hallucinations or confabulations happen when models perceive patterns in their training data sets that do not exist in the real world.

For example, prompting ChatGPT-4o mini with the question “In which episode of *Seinfeld* does Kramer go to Bangladesh?” returned this answer:

Kramer travels to Bangladesh in the episode titled “The Little Jerry,” which is the 145th episode of Seinfeld. It is the 145th [sic] episode of the show, and it originally aired during Season 8. In this episode, Kramer becomes involved in a scheme to import a toy from Bangladesh.

While *The Little Jerry* is the 145th episode of *Seinfeld* and aired during Season 8, it features Kramer buying and training a rooster—the eponymous Little Jerry.

Prompting the chatbot again with the same question returned a second, different hallucination:

Kramer goes to Bangladesh in the episode titled "The Maldives," which is the 18th episode of Season 9. In this episode, he tries to start a new life after his trip, and the storyline revolves around the idea of him becoming a "Bangladeshi" fisherman.

In reality, episode 18 of season 9 of *Seinfeld* is *The Frogger*, and in it, Kramer only goes so far as the police station. There is no episode of *Seinfeld* called *The Maldives*, and at no point during the show’s entire run does Kramer go to Bangladesh.

Just as autoregressive models such as GPT are known to hallucinate with made-up facts, other types of models can hallucinate in their own respective ways. Diffusion models used for image generation sometimes depict real-world objects in ways that do not match reality.

How to mitigate hallucinations

While it’s not yet possible to entirely remove the possibility for generative AI models to hallucinate, best practices for hallucination mitigation include:

- **Clear prompting:** The more explicit the user prompt, the more focused the AI’s answer can be. Give the AI space within the prompt to answer with detail.
- **Focused direction:** Giving an AI a clear role and instructing it to provide truthful, verifiable information can help ensure its answers better reflect reality.
- **High-quality data:** The more current and relevant an AI model’s training data, the lower the chance that its answers will be biased.
- **Human verification:** AI-generated results should not be used without being first verified by knowledgeable humans.

- **RAG and fine-tuning:** Using RAG to augment an AI with credible data and fine-tuning models to become more domain-specific are both effective in reducing hallucinations.

Misinformation and disinformation and deep fakes

Misinformation and disinformation are commonly known as ‘fake news’. The mis/disinformation phenomenon has been catalysed by the technological revolution. The increasingly frequent use of social media contributes to the amount of data shared and processed by algorithms. It has also changed the way news is created and disseminated.

Put simply, misinformation is the sharing of incorrect information whereas disinformation is the sharing of incorrect information with a malicious intent. Mis/disinformation can take many forms such as memes, misattributed content, fabricated or cloned websites and news. It also includes altered and fabricated images and videos (these are also called deepfakes). Bots, trolls and other AI or non-AI agents can increase the reach of mis/disinformation campaigns. Algorithms that create filter bubbles significantly intensify its impact and effect on users.

Misinformation and disinformation and deep fakes and human rights

We assert that misinformation and disinformation amplifies discrimination and human bias, having an adverse effect on minority groups that are already discriminated against. There are also other human rights that may be affected.

The impact on human rights during elections

Article 25 the International Convention on Civil and Political Rights (ICCPR) establishes the right of everyone to take part in the conduct of public affairs and to vote. Mis/disinformation campaigns have been largely used during elections or referenda to manipulate voters and to shape the outcome of polls. Bots have been used to affect the public debate and influence the electorate both in domestic and international elections.

In Ecuador and in the Philippines the presidents admitted using paid trolls during the electoral campaigns . Studies also confirmed the use of state-led propaganda by the Russian Internet Research Agency as a foreign policy tactic to affect the outcome of the 2016 US Presidential Elections. Misleading and

deceiving information was provided to targeted groups to undermine their right to vote. Similar campaigns have also been used to discredit electoral observation missions.

The threats to human rights defenders and journalists and the chilling effect

Computational propaganda' can also be used to target or troll journalists and human rights defenders critical of governments or political movements. This can include formal state sponsorship and other organisational sponsors. The goal of such targeting is to create reputational damage, a societal backlash, or a chilling effect that discourages the continuation of critical work, as warned by the UN Secretary-General in his opening remarks to the Human Rights Council in February 2019. These online harassment campaigns can produce a chilling effect on the freedom of expression, association and assembly of the targeted individuals, who may refrain from publicly expressing their views and engaging in their normal activities for fears of further verbal and physical attacks.

Hiding behind mis/disinformation and avoiding accountability

Governments around the world have also started using the label 'fake news' or 'mis/disinformation' to close down debates or deny allegations against them in the context of human rights violations. This strategy represses freedom of expression and avoids accountability, threatening many human rights guarantees such as the duty to investigate in the case of violation of the right to life.

Mis/disinformation as a tool for hate speech

A significant proportion of mis/disinformation, and memes in particular, contain hate speech: incitement to hatred, justification for and promotion of spreading hateful messages based on intolerance, racism and violence. This is boosted by the fact that the digital space offers the possibility to post anonymously, resulting in a lack of accountability.

The responses to mis/disinformation and their threats to human rights

States, business and international organisations have been developing responses to tackling mis/disinformation, ranging from establishing expert groups and task forces, demotion/removal of content and accounts, anti-disinformation laws and media literacy programmes. Some of these measures may pose further risks to human rights, in particular to freedom of expression.

Definitions

Fake news: “Fictions deliberately fabricated and presented as non-fiction with the intent to mislead recipients into treating fiction as fact or into doubting verifiable fact.” Eric Emanuelson, Jr., Fake Left, Fake Right: Promoting an Informed Public in the Era of Alternative Facts, 70 Admin. L. Rev. 209, 218-19 (2018).

Deepfakes: Media created with machine learning and/or artificial intelligence in which a person in an existing image or video is replaced with someone else’s likeness. Viorela Dan et al., Visual Mis- and Disinformation, Social Media, and Democracy, 98 Journalism & Mass Communication Quarterly 641, 643 (2021).

Misinformation: A “verifiably false statement of fact.” Charles R. Corbett, Chemtrails and Solar Geoengineers: Governing Online Conspiracy Theory Misinformation, 85 Mo. L. Rev. 633, 652 (2020).

Disinformation: “[F]alse information that is purposely spread to deceive people.” David M. J. Lazer et al., The science of fake news, 359 Science 1094 (2018).

Information literacy: A set of “competencies that enable people to critically and effectively engage with communications content; the institutions that facilitate this content; and the use of digital technologies.” Media and information literate citizens: think critically, click wisely!, UNESCO Digital Library.

Psychological Operation (PSYOP): “Often confused with Information Operations as a whole, PSYOP refers to influence activities specifically intended ‘to induce or reinforce foreign attitudes and behavior in a manner favorable to U.S. objectives.’” 12 (March 8, 2011) Terrorist Use of the Internet: Information Operations in Cyberspace.

DEEPPFAKE

In the opening session of his 2020 introductory course on deep learning, Alexander Amini, a PhD student at the Massachusetts Institute of Technology (MIT), invited a famous guest: former US President Barack Obama.

“Deep learning is revolutionizing so many fields, from robotics to medicine and everything in between,” said Obama, who joined the class by video conference.

After speaking a bit more on the virtues of artificial intelligence, Obama made an important revelation: “In fact, this entire speech and video are not real and were created using deep learning and artificial intelligence.”

Watch this video: <https://youtu.be/l82PxsKHxYc>

Amini’s Obama video was, in fact, a *deepfake*—an AI-doctored video in which the facial movements of an actor are transferred to that of a target. Since first appearing in 2018, deepfake technology has evolved from hobbyist experimentation to an effective and dangerous tool. Deepfakes have been used against celebrities and politicians and have become a threat to the very fabric of truth.

How Do Deepfakes Work?

Deepfake applications work in various ways. Some transfer the facial movements of an actor to a target video, such as the one we saw at the beginning of this article, or this Obama deepfake created by comedian Jordan Peele to warn about the threat of fake news:

Watch this video : <https://youtu.be/cQ54GDm1eL0>

Other deepfakes map the face of a target person onto other videos—for example, this video of Nicolas Cage’s face mapped onto that of characters in different movies.

Watch this video : <https://youtu.be/BU9YAHigNx8>

Like most contemporary AI-based applications, deepfakes use deep neural networks (that’s where the “deep” in deepfake comes from), a type of AI algorithm that is especially good at finding patterns and correlations in large sets of data. Neural networks have proven to be especially good at computer vision, the branch of computer science and AI that handles visual data.

Deepfakes use a special type of neural-network structure called an “autoencoder.” Autoencoders are composed of two parts: an encoder, which compresses an image into a small amount of data; and a decoder, which decompresses the compressed data back into the original image. The mechanism is similar to those of image and video codecs such as JPEG and MPEG.

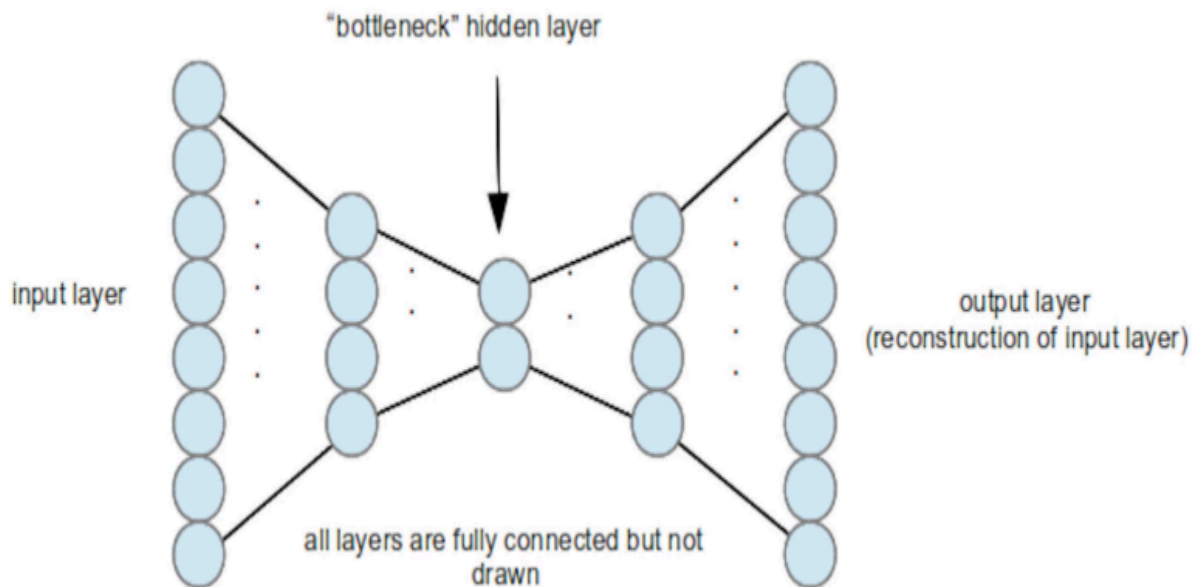


Image courtesy of Two Minute Papers

But unlike classical encoder/decoder software, which work on groups of pixels, the autoencoder operates on the features found in images, such as shapes, objects, and textures. A well-trained autoencoder can go beyond compression and decompression and perform other tasks—say, generating new images or removing noise from grainy images. When trained on images of faces, an autoencoder learns the features of the face: the eyes, nose, mouth, eyebrows, and so on.

Deepfake applications use two autoencoders—one trained on the face of the actor and the other trained on the face of the target. The application swaps the inputs and outputs of the two autoencoders to transfer the facial movements of the actor to the target.

creating deepfakes is neither trivial nor fully automated. The technology is gradually getting better, but creating a decent deepfake still requires a lot of time and manual work.

1. First, you have to gather many photos of the faces of the target and the actor, and those photos must show each face from different angles. The process usually involves grabbing thousands of frames from videos that feature the target and actor and cropping them to contain only the faces. New deepfake tools such as Faceswap can do part of the legwork by automating the frame extraction and cropping, but they still require manual tweaking.
2. Training the AI model and creating the deepfake can take anywhere from several days to two weeks, depending on your hardware configuration and the quality of your training data.

The Fight against Deepfakes

Earlier deepfakes contained visual artifacts that were visible to the naked eye, including unnatural eye blinking and abnormal skin color variations. But deepfakes are constantly improving.

Researchers have been devising new techniques to detect deepfakes only to see them become ineffective as the technology continues to evolve and yield more natural results. So as the 2020 presidential elections close in, major tech companies and government agencies have been racing to counter the spread of deepfakes.

In September, Facebook, Microsoft and several universities launched a competition to develop tools that can detect deepfakes and other AI-doctored videos. “This is a constantly evolving problem, much like spam or other adversarial challenges, and our hope is that by helping the industry and AI community come together we can make faster progress,” Facebook CTO Michael Schroepfer wrote in a blog post that introduced the Deepfake Detection Challenge. The social media giant has allocated \$10 million to the industry-wide effort.

DARPA, the research arm of the Department of Defense, has also launched an initiative to curb the spread of deepfakes and other automated disinformation attacks. In addition to detecting doctored videos and images, DARPA will be looking for ways to facilitate attribution and identification of the parties involved in the creation of fake media.

Other efforts at universities and research labs range from using deep learning to detect modified areas in images to using blockchain to establish a ground truth and register trustable videos.

But all in all, researchers agree that the fight against deepfakes has become a cat-and-mouse chase. As one researcher told me last year, “Whatever we do, people who create those manipulations come up with something else. I don't know if there will ever be a time where we will be able to detect every kind of manipulation.”

Deepfakes and the Technology Behind It from BBC Click

<https://youtu.be/8LhI-e2B8Lg>

How To Spot A Deepfake Like The Barack Obama–Jordan Peele Video

The Obama Peele video was created using another emerging technology: FakeApp. It's a free tool that's recently been used to insert the faces of celebrities into porn videos. Other people have put Nic Cage into Indiana Jones, face-swapped Nic Cage and Lois Lane, and created other cinematic crimes against humanity.

Part of the process for creating this video involved taking an original video of Barack Obama and pasting Jordan Peele's mouth into it. It looked really bad and clumsy at first. But things got remarkably better the longer FakeApp was left to process a more believable amalgamation of Obama's head and Peele's mouth. That took roughly 56 hours and was overseen by a video effects professional.

So the good news is it still requires a decent amount of skill, processing power, and time to create a really good “deepfake.”

There are basic tips you can follow to ensure you don't get fooled easily. If more people are equipped to spot a fake, then the world has a better shot at avoiding a dystopian hellscape where every movie stars Nic Cage, and governments, trolls, and companies manufacture perfect fakes that collectively make us lose touch with reality.

1. Don't jump to conclusions

One general best practice is to “refrain from jumping to strong conclusions in either direction without additional information and confirmation,” said Hany Farid, a professor of computer science at Dartmouth College and one of the world's leading image forensics experts. Wait and see what other information comes to light about the video or piece of content in question, and hold back on sharing it with others. This is especially true in breaking news situations, where information moves quickly and is often wrong or misinterpreted in the early hours.

2. Consider the source

Pay attention to where this piece of content originated. Was it uploaded to a random account on a social network? Who is claiming ownership of it, and where do they say it came from?

3. Check where else it is (and isn't) online

Do an online search with the video's title or other relevant keywords to see if multiple credible news outlets and the person or entity in question have weighed in on it. Again, no need to rush to a conclusion until you can consult multiple courses.

4. Inspect the mouth

Farid said that today's deepfake videos often have telltale signs. One thing to look closely at is the mouth of the person speaking, because these tools often struggle “to accurately render the teeth, tongue, and mouth interior. So, keep a watchful eye on the mouth for [visual anomalies].”

5. Slow it down

Farid also encourages people to slow down and freeze parts of the video to watch it more closely. “Slowing a video down so that you can clearly see the frame-to-frame transition can sometimes reveal temporal glitches that are introduced in manipulated video,” he said.

Watch this video :

https://video-player.buzzfeed.com/embed?video_id=54375